Does temporal information need a learned gate?

DS7400 HW3 | Controlled extension of the HW2 environmental-sound MLP

1. Original hypothesis and design rationale

HW2 reduces a recording to whole-clip means and standard deviations, losing within-clip temporal structure. The hypothesis, committed before training, was that four ordered segments would preserve short events, and an input-dependent gate could suppress unreliable local evidence. Expected risks were overfitting, sigmoid saturation, fixed-boundary sensitivity, and padding in short clips. This is a task-specific application of established gating principles, not a claim to invent gating.

2. Dataset and controlled experiments

UrbanSound8K contains 8,732 clips in 10 classes (36.4 times the HW2 sample count). The task remains environmental-sound classification, but the labels differ: HW2 accuracy is not directly comparable. The baseline retrains the HW2 architecture with a 10-class head: 168 global features, 64 ReLU units, dropout 0.5, and cross-entropy loss. Audio is mono, 22,050 Hz, padded to 5 seconds; 64 log-Mel and 20 MFCC temporal means/stds form the 168 features. All official short clips are retained.

For each of ten official test folds, the next cyclic fold validates and the other eight train. An audit found five source IDs across folds (152 clips). Test sources are purged from validation; test and retained validation sources are purged from training. All test clips remain. Scaling uses training data only. All four models train for 80 epochs with SGD (LR 0.03, momentum 0.9, weight decay 0.001, batch 128); the lowest-validation-loss checkpoint is restored. Seeds are paired across models (7400 + fold). This is ten-fold outer evaluation with separate validation, not the published protocol that trains on all nine non-test folds. No test-driven tuning was performed.

Proposed mapping: global 168 -> 64 ReLU = g; local 672 -> 64 ReLU = l; a = sigmoid(Linear([g;l])); h = (1-a)g + a·l; dropout(h) -> 10 logits. The architecture diagram is also included in the repository.

Controls: concatenation feeds all 840 features to one 64-unit layer; fixed fusion uses the same two branches with a = 0.5. Fixed and gated start from identical branch/head weights, batch order and dropout stream. Only 129 gate parameters differ. The baseline is much smaller, so gated-versus-fixed is the cleaner test of routing.

3. Observed results: all ten test folds

Model	Accuracy ± fold SD	Macro F1	Parameters	Seconds/run
HW2 MLP	57.86% ± 5.71	0.589	11,466	2.3
Concatenation	60.46% ± 4.11	0.610	54,474	3.3
Fixed fusion	63.81% ± 5.23	0.651	54,538	3.9
Learned gate	61.81% ± 4.67	0.625	54,667	4.3

Accuracy and F1 are unweighted means over folds. SD describes partition variability, not a confidence interval. Timing includes training/evaluation with four fold workers sharing the CPU; it is not an isolated speed benchmark.

Analysis: what the experiments actually support

4. Compare observations with the hypothesis

The learned gate differs from **HW2 MLP** by **+3.95 percentage points** on average, with higher accuracy in 9 of 10 folds. The fold-paired difference SD is 3.15 points.

The learned gate differs from **Fixed fusion** by **-2.01 percentage points** on average, with higher accuracy in 2 of 10 folds. The fold-paired difference SD is 3.01 points.

The learned gate differs from **Concatenation** by **+1.35 percentage points** on average, with higher accuracy in 5 of 10 folds. The fold-paired difference SD is 4.05 points.

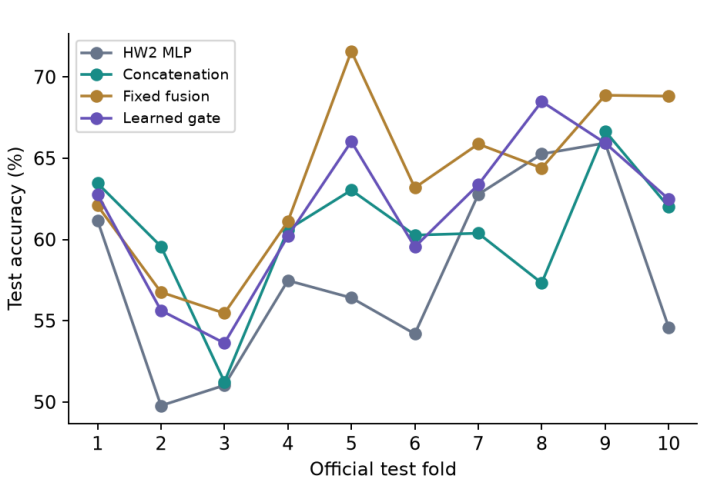
The proposed gate is not the strongest design: Fixed fusion has the highest mean accuracy. A richer temporal representation and a learned routing mechanism are separate claims. The concat and fixed controls prevent attributing every change from the small baseline to the gate. These measurements do not establish that input-dependent gating is necessary.

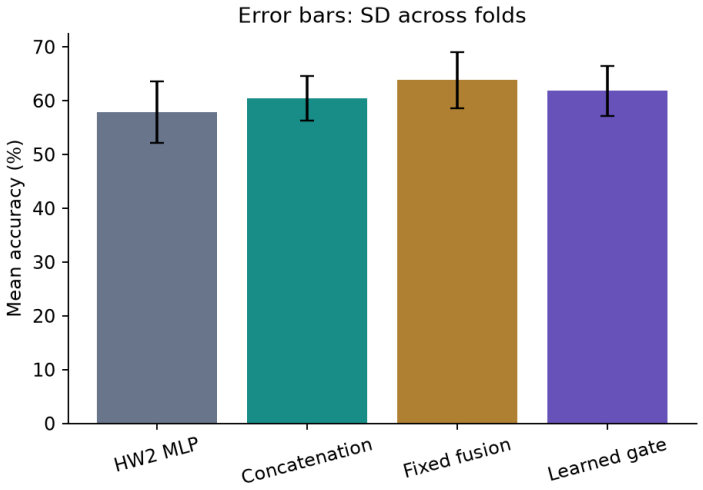
5. Mechanism, failures and alternative explanations

The local-branch gate has pooled mean 0.767, SD 0.319, and range 0.000-1.000. 49.9% of clips have gates below 0.05 or above 0.95. This substantial saturation is consistent with hard routing and small sigmoid derivatives, a plausible failure mechanism rather than causal proof. These coefficients are not feature importance. Class-dependent weighting can also reflect padding or background conditions.

Reversing segment order at evaluation changes gated accuracy from 61.81% to 31.99%. This diagnostic is deliberately excluded from model selection. It changes positional distributions, including padding and position-specific scaling, so a drop is not conclusive evidence of meaningful chronological reasoning.

Mean gated training cross-entropy falls from 1.102 after epoch 1 to 0.181 after epoch 80. Selected-checkpoint train/validation accuracy averages 79.22%/62.26%; this gap suggests generalization limitations. No nonfinite-loss failure occurred. Best-checkpoint selection limits over-training, but does not remove dataset shift or overfitting.





6. What was learned and what comes next

Extra temporal information, branch structure and adaptive routing must be tested separately. More complexity is not automatically better. Next, repeat seeds, add a parameter-matched global MLP, and test a gentler gate (temperature or regularization) plus valid-duration segments. These distinguish capacity, saturation and padding effects. Choose those settings using validation data, without revisiting this test comparison as a tuning target. The repository includes every history, out-of-fold prediction, checkpoint, executed notebook and a reproducible config.

Reference: J. Salamon, C. Jacoby & J. P. Bello, A Dataset and Taxonomy for Urban Sound Research, ACM Multimedia (2014). Dataset: doi.org/10.5281/zenodo.1203745. AI assistance: brainstorming, implementation, debugging and document preparation; design direction selected by the user.